Problem 1. Let G and J be groups and let $\varphi : G \to H$ be a group epimorphism (surjective homomorphism)

(a) Prove that if J is abelian, then any subgroup of G that contains $\ker(\varphi)$ is a normal subgroup of G.

Proof. Note that the First Isomorphism Theorem says that

$$G/\ker(\varphi) \cong \varphi(G)$$

Furthermore, $\varphi(G) = J$ since φ is surjective. So, $G/\ker(\varphi) \cong J$ and hence, $G/\ker(\varphi)$ is abelian. Since J is abelian, every subgroup of $G/\ker(\varphi)$ is normal. Hence, by the 4th Isomorphism Theorem, every subgroup of G containing $\ker(\varphi)$ is also normal. ■

(b) Suppose $K \trianglelefteq J$. Prove that there exists $H \trianglelefteq G$ such that $\ker(\varphi) \subseteq H$ and G/H is isomorphic to J/K.

Proof. Suppose $K \trianglelefteq J$. We will prove that our candidate for H will be $\varphi^{-1}(K) \leq G$. Note that $\varphi^{-1}(K) \leq G$ by problem 1 of homework 2. Since J is abelian, it follows from part (a) that $\varphi^{-1}(K) \trianglelefteq G$ and contains $\ker(\phi)$. Hence, we can see that our candidate must be $H = \varphi^{-1}(K)$. Indeed, w can see that this is the case through the First Isomorphism Theorem. That is, let us define the mapping

$$\psi : G \to J/K \text{ by } g \mapsto \varphi(g)K.$$

Note that this is a homomorphism because for all $g_1, g_2 \in G$, we have (coset multiplication is well-defined since $K \trianglelefteq G$ and the fact that φ is a homomorphism)

$$\begin{aligned}\psi(g_1g_2) = \varphi(g_1g_2)K &= \Big(\varphi(g_1)\varphi(g_2)\Big)K \\ &= \varphi(g_1)K\varphi(g_2)K \\ &= \psi(g_1)\psi(g_2).\end{aligned}$$

Also, we have

$$\begin{aligned}\ker(\psi) &= \{g \in G : \psi(g) = 1_H K\} \\ &= \{g \in G : \varphi(g)K = 1_H K\} \\ &= \{g \in G : \psi(g) \in K\} \\ &= \varphi^{-1}(K).\end{aligned}$$

Hence, we have $\ker(\psi) = \varphi^{-1}(K) = H$. Because φ is a surjective map, it also follows from our definition of ψ that ψ must also be surjective. Indeed, for any $h \in H$, there exists an $g \in G$ such that $\varphi(g) = h$. Hence, we have

$$hK = \varphi(g)K = \psi(g)$$

and so $\psi(G) = J/K$. Using the First Isomorphism Theorem, we can see that

$$G/\ker(\psi) \cong \psi(G) \iff G/H \cong J/K$$

where $H = \varphi^{-1}(K)$.

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Problem 2. Suppose that $\pi : \mathbb{Z}_{30} \to G$ is an onto homomorphism where G is a group of order 5. Determine, with proof, the kernel of φ. ■

Proof. Observe, by the First Isomorphism Theorem, (because π is an onto homomorphism) that

$$\mathbb{Z}_{30}/\ker(\pi) \cong \pi(\mathbb{Z}_{30}) \iff \mathbb{Z}_{30}/\ker(\pi) \cong G.$$

This tells us that, since G and $\mathbb{Z}_{30}$ are finite groups, it follows that

$$|\mathbb{Z}_{30}/\ker(\pi)| = |G| \iff \frac{|\mathbb{Z}_{30}|}{|\ker(\pi)|} = |G|.$$

Since $|G| = 5$ and $|\mathbb{Z}_{30}| = 30$, we have from the above that $|\ker(\varphi)| = 6$. This tells us that there are six elements of $\mathbb{Z}_{30}$ that gets mapped to the identity 1_G. Since the divisors of 30 gives us unique cyclic subgroups of $\mathbb{Z}_{30}$, all we need to do is find the subgroup of the same order as $\ker(\varphi)$. Indeed, this subgroup of order 6 must be the cyclic subgroup generated by $\overline{5} \in \mathbb{Z}_{30}$. That is, $\ker(\pi) = \langle \overline{5} \rangle = \{\overline{0}, \overline{5}, \overline{10}, \overline{15}, \overline{20}, \overline{25}\}$. ■

Problem 3. Suppose that $|G| = p^2 q$ where p and q are primes with $p > q$. Show that if G has a subgroup H of order p^2, then H is the unique subgroup of order p^2 in G. Deduce that G/H is cyclic. ■

Proof. Suppose that $|G| = p^2 q$ where p and q are primes with $p > q$. Suppose G has a subgroup H of order p^2. Suppose there exists another subgroup $K \leq G$ such that K is order p^2. Since H and K are finite, it follows that

$$|HK| = \frac{|H||K|}{|H \cap K|} = \frac{p^2 \cdot p^2}{|H \cap K|} = \frac{p^4}{|H \cap K|}.$$

This tells us that $|H \cap K| \mid p^4$ and so we have; that is,

$$|H \cap K| \in \{1, p, p^2, p^3, p^4\}.$$

If $|H \cap K| = 1$, then $|HK| = p^4$. Since $p > q$, it follows that

$$|HK| = p^4 > p^3 q > p^2 q = |G|.$$

But this is impossible since $HK \subseteq G$. Hence, $|H \cap K| \neq 1$.

If $|H \cap K| = p$, then

$$|HK| = \frac{p^4}{p} = p^3 \implies |HK| = p^3 > p^2 q = |G|$$

and again, we have the same situation as the last case. Hence, $|H \cap K| \neq p$.

If $|H \cap K| = p^2$, then $|H \cap K| = |H| = |K|$. Since $H \cap K \leq H \leq G$ and $H \cap K \leq K \leq G$, we know that

$$H = H \cap K = K \implies H = K.$$

Hence, H is unique.

If $|H \cap K| = p^3$, then $|HK| = \frac{p^4}{p^3} = p < p^2 = |H| = |K|$. But this tells us that HK contains less number of elements than H and K which is absurd.

If $|H \cap K| = p^4$, then $|HK| = \frac{p^4}{p^4} = 1 < p^4 = |H||K|$. But this is also absurd because the number of elements in HK must contain more than 1 element since H and K are non-trivial subgroups of G. Thus, from the above we can say that we must have $|H \cap K| = p^2$. Hence, $H = K$ and so H is a unique subgroup of G of order p^2. ■

In the following problem, we will use the fact that

$$\text{lcm}(m, n) = \frac{mn}{\gcd(m, n)}.$$

Problem 4. Suppose that H and K are subgroups of finite index in a possibly infinite group G with

$|G : H| = m$ and $|G : K| = n$. Prove that

$$\text{lcm}(m, n) \leq |G : H \cap K| \leq mn$$

. Deduce that $|G : H \cap K| = mn$ when m and n are relatively prime. ■

Proof. Suppose $|G : H| = m$ and $|G : K| = n$. To show the above result, it suffices to show that

$$1 \leq |G : H \cap K| \leq \frac{mn}{\text{lcm}(m, n)}. \tag{*}$$

That is,

$$1 \leq |G : H \cap K| \leq \gcd(m, n).$$

Since $H \cap K \leq H \leq G$ and $H \cap K \leq K \leq G$, we have

$$\begin{aligned} m = |G : H| = |G : H \cap K||H \cap K : H| \\ n = |G : K| = |G : H \cap K||H \cap K : K|. \end{aligned}$$

This tells us that $|G : G \cap K| \mid m$ and $|G \cap K| \mid n$. By definition of gcd, it follows that $1 \leq |G : G \cap K| \leq \gcd(m, n)$. That is,

$$1 \leq |G : H \cap K| \leq \frac{mn}{\text{lcm}(m, n)}.$$

Hence,

$$\text{lcm}(m, n) \leq |G : H \cap K| \leq mn,$$

proving (*). ■

Problem 5. Prove Theorem 16 and corollary 17 from section 3.3 of the textbook. ■

Theorem (The First Isomorphism Theorem)**.** If $\varphi : G \to H$ is a homomorphism of groups, then $\ker(\varphi) \trianglelefteq G$ and $G/\ker(\varphi) \cong \varphi(G)$.

Proof. Define $\mu : G/\ker(\varphi) \to \varphi(G)$ by $\mu(gK) = \varphi(g)$ where $K = \ker(\varphi)$. To see that K is normal, let $g \in G$ and let $y \in gKg^{-1}$. Then $y = gkg^{-1}$ for some $k \in \ker(\varphi)$. Then $\varphi(k) = 1_H$ and so

$$\begin{aligned} \varphi(y) = \varphi(gkg^{-1}) &= \varphi(g)\varphi(k)\varphi(g^{-1}) \\ &= \varphi(g)\varphi(k)(\varphi(g))^{-1} \\ &= \varphi(g)1_H(\varphi(g))^{-1} \\ &= \varphi(g)(\varphi(g))^{-1} \\ &= 1_H. \end{aligned}$$

Hence, it follows that $y \in \ker(\varphi)$ and so, $K = \ker(\varphi)$ must be a normal subgroup. In what follows, we will show that μ isomorphism. First, we want to make sure that μ is well-defined. To this end, suppose $g_1K = g_2K$ where $g_1, g_2 \in G$. This is true if and only if $g_1^{-1}g_2 \in \ker(\varphi)$. This implies that $\varphi(g_1^{-1})\varphi(g_2) = 1_G$ which is true if and only if

$$(\varphi(g_1))^{-1}\varphi(g_2) = 1_G \iff \varphi(g_2) = \varphi(g_1) \iff \mu(g_2K) = \mu(g_1K).$$

This tells us that μ is well-defined and also that μ is injective. Next, we will show that μ is a homomorphism. Indeed, for all $g_1K, g_2K \in G/K$, we have that (since K is a normal subgroup)

$$\begin{aligned}
\mu(g_1Kg_2K) &= \mu(g_1g_2K) \\
&= \varphi(g_1g_2) \\
&= \big(\varphi(g_1)\varphi(g_2)\big) \\
&= \varphi(g_1)\varphi(g_2) \\
&= \mu(g_1K)\mu(g_2K).
\end{aligned}$$

Lastly, we show that μ is surjective. It is clear from the definition that this is the case. Indeed, for any $y \in \varphi(g)$, it follows that $\varphi(g) = y$ for some $g \in G$ and so we have

$$\mu(gK) = \varphi(g) = h.$$

Hence, we see from the above argument that μ must be an isomorphism and so we conclude that

$$G/\ker(\varphi) \cong \varphi(G).$$

■

Corollary (17). Let $\varphi : G \to H$ be a homomorphism of groups. Then

(1) φ is injective if and only if $\ker(\varphi) = 1$.

(2) $|G : \ker(\varphi)| = |\varphi(G)|$.

Proof. First, we show (1). Since $\varphi : G \to H$ is a homomorphism, it follows from the First Isomorphism Theorem that

$$G/\ker(\varphi) \cong \varphi(G).$$

If $\ker(\varphi) = \{1_G\}$, then

$$G/\ker(\varphi) = G/\{1_G\} \cong G.$$

This holds if and only if

$$G \cong \varphi(G).$$

which is true if and only if φ is injective. Next, we show (2). Indeed, we can see from the First Isomorphism Theorem that (since φ is a homomorphism)

$$\begin{aligned}
G/\ker(\varphi) \cong \varphi(G) &\iff |G/\ker(\varphi)| = |\varphi(G)| \\
&\iff |G : \ker(\varphi)| = |\varphi(G)|.
\end{aligned}$$

■

Problem 6. Let G be a group with a normal subgroup N. Prove that there is a bijection from the subgroups of G containing N onto the subgroup of G/N. ■

Proof. Denote the set of all subgroups of G containing N as

$$\mathcal{A} = \{K \leq G : N \leq K,\ N \trianglelefteq G\}$$

and

$$\mathcal{B} = \{\overline{K} = K/N : \overline{K} \leq G/N\}.$$

Define the mapping $\Phi : \mathcal{A} \to \mathcal{B}$ by $\Phi(K) = \overline{K}$. We will show that Φ is bijective so that there is a one-to-one correspondence between $\mathcal{A}$ and $\mathcal{B}$. First, we will show that Φ is injective. Suppose that

$\Phi(K_1) = \Phi(K_2)$. Then $\overline{K_1} = \overline{K_2}$; that is, $K_1/N = K_2/N$. Our goal is to show that $K_1 = K_2$. To show this, we will show that $K_1 \leq K_2$ and $K_2 \leq K_1$. Here we will show the first containment as the other containment can be obtained using a similar argument. Let $k_1 \in K_1$. Then consider the coset k_1N. Since $\overline{K_1} = \overline{K_2}$, it follows that $k_1N \in \overline{K_2}$. That is, there exists a $k_2 \in K_2$ such that

$$k_1N = k_2N \Longrightarrow k_1^{-1}k_2 \in N.$$

Since $N \leq K_2$, it follows that $k_1^{-1}k_2 \in K_2$. Then for some $\tilde{k}_2 \in K_2$, we have $\tilde{k}_2 = k_1^{-1}k_2$ and so

$$k_1 = k_2\tilde{k}_2^{-1} \Longrightarrow k_1 \in K_2.$$

by the subgroup criterion. Hence, $K_1 \leq K_2$ and with a similar argument with just the roles of k_1 and k_2 reversed, it follows that $K_2 \leq K_1$.

Hence, we see that $K_1 = K_2$.

Lastly, we show that Φ is surjective. Let $\overline{J} \leq G/N$. Consider the projection map $\pi : J \to J/N$. The complete pre-image of $\overline{J}$ under π is $\pi^{-1}(\overline{J})$ which is a subgroup of G which also contains N. Hence, $\pi^{-1}(\overline{J}) = K \leq G$ and so $\Phi(K) = \overline{J}$ and so Φ is surjective.

Since Φ is injective and surjective, it follows that Φ is bijective and hence we have our one-to-one correspondence. ■